

9

MAIN MEMORY

9.1. Introduction

- The main memory is the central storage unit in a computer system. The principal technology used for the main memory is based on semiconductor integrated circuits. Main memory is made up of RAM and ROM chips. Different types of integrated circuit memories are given in Table 1.
- Integrated circuit RAM chips are available in two possible operating modes, static and dynamic. The static RAM consists of interval flip flops that store the binary information. The stored information remains valid as long as power is applied to the unit i.e. ROM is volatile. The dynamic RAM stores the binary information in the form of electric charges that are applied to capacitors.

Memory	Access	Volatile	Features
Read/Write	Random or serial	Yes	Data can be read or written with equal ease. Used whenever data is needed fast and often, and is frequently changed.
Read-only (ROM)	Random	No	(a) Data is stored permanently by a masking step during manufacture. (b) Used whenever data is not to be changed as in fixed tables, constants or computer instruction sets.
Programmable Read-only (PROM)	Random	No	(a) Can be programmed after IC is manufactured. (b) Data is written by opening fusible metal links or P-N junction's with high currents. (c) Can only be written into once. Same use as a ROM, but is cheaper in small quantities.
Re-programmable read-only	Random	No	Are ROMs that can be written into many times. Are really "read-mostly" rather than "read-only" memories Two main types: (a) Those in which writing is by voltage application and erasing (all data at once) by exposing chip to ultra-violet radiation through a window on the IC (b) Those in which reading and writing is electrical. Data remains stored even with no power applied through the use of MNOS transistors. Used where frequent or at least more than one change in a program is required as in debugging a program.
Content-addressable (CAM)	Random	Yes	This extracts all data in an address when a part of the contents of that address match a specified number. Used in associative memories to obtain stored data related in some way to the input data.

Charge-coupled device (CCD)	Serial	Yes	(a) Digital input is converted to charge and stepped through a shift register. (b) Requires refresh circuitry to prevent data loss.
Programmable logic array (PLA)	Random	No	(a) Is a memory structure that is mask or field (FPLA) programmed as is a ROM. However, it implements logic function. (b) Inputs are functions of the input variables and the logic operation stored in the address. (c) Is more flexible than random (hard-wired separate IC) circuits because PLAs and FPLAs are programmable.

Refreshing is done by cycling through the words every few milliseconds to restore the decaying charge. The dynamic RAM offers reduced power consumption and larger storage capacity in a single memory chip. The static RAM is easier to use and has shorter read and write cycles. The Table 2 shows the summary of various types of integrated circuit memories.

(a) Types of Memory Access	
Random access	Any address can be accessed at random, that is, without going through other address first. Data retrieval time is relatively fixed.
Serial access	Typified by shift register or charge-coupled-device (CCD) memories. Data retrieval is serial in a fixed order. All data ahead of required data must be read first.
(b) Static versus Dynamic Storage	
Static storage	Data is stored in flip-flops or other memory cells in which the data does not deteriorate with time.
Dynamic storage	Data is stored in leaky capacitors so that refresh circuitry is required to prevent data loss.

9.2. Memory Interface

A computer uses memory capacity of 512 bytes of RAM and 512 bytes of ROM. The IC sizes of RAM and ROM are 128×8 and 512×8 bits respectively.

- Find the number of RAM ICs.
- Find the number of ROM ICs.
- Give the memory map of the system
- Mention the size of decoder.
- Give the diagram of memory connection to the CPU.

Solution:

$$\begin{aligned}
 \text{(a) The number of RAM ICs} &= \frac{\text{Total RAM size}}{\text{RAM IC size}} \\
 &= \frac{512 \times 8}{128 \times 8} \\
 &= 4
 \end{aligned}$$

(b) The number of ROM ICs = $\frac{\text{Total ROM size}}{\text{ROM IC size}}$

$$= \frac{512 \times 8}{512 \times 8}$$

$$= 1$$

(c) The memory map of this system is given by Table. 3

Component	Hexadecimal address	Address bus											
		10	9		8	7	6	5		4	3	2	1
RAM 1	0000–007F	0	0		0	×	×	×		×	×	×	×
RAM 2	0030–00FF	0	0		1	×	×	×		×	×	×	×
RAM 3	0100–017F	0	1		0	×	×	×		×	×	×	×
RAM 4	0180–01FF	0	1		1	×	×	×		×	×	×	×
ROM	0200–03FF	1	×		×	×	×	×		×	×	×	×

(d) 2×4 decoder

Each RAM receives the seven low-order bits of the address bus to select one of 128 possible bytes. The particular RAM chip selected is determined from lines 8 and 9 in the address bus. This is done through a 2×4 decoder. The outputs of decoder are given to the CSI inputs in each RAM chip. Thus when address lines 8 and 9 are equal to 00, the first RAM chip is selected when 01, the second RAM chip is selected and so on.

(e) The connection of memory chips to the CPU is shown in Fig. RAM and ROM chips are connected to a CPU through the data and address buses. The low order lines in the address bus selected the byte within the chips and other lines in the address bus select a particular chip through its chip select inputs.

